Expectation Maximization Algorithm (A soft clustering method)

Expectation-Maximization

- The Expectation–Maximization (EM) algorithm is a method used to learn parameters of a model when some data is missing or hidden (latent variables).
- **Real-Life Intuition**
- Imagine:
  - You have exam scores from two classes mixed together
  - You don’t know which student belongs to which class
- EM will:
  - Guess class membership (E-step)
  - Recompute class averages (M-step)
  - Repeat until separation becomes clear

Latent variables : variables that are not directly observable and are actually inferred from the values of the other observed variables

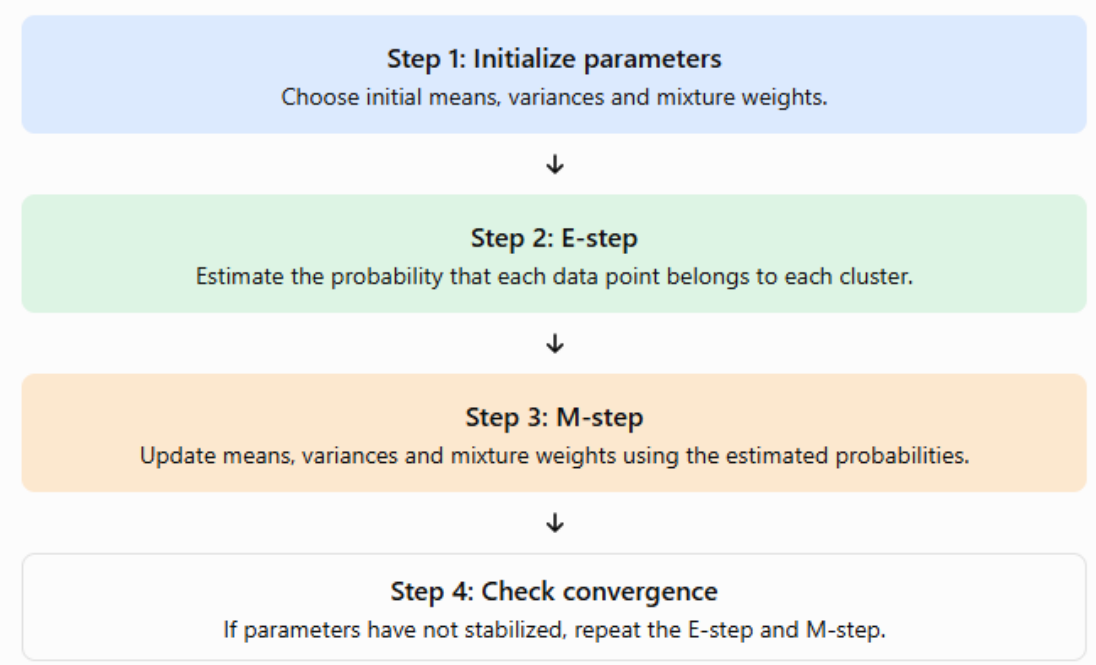
Key Terms

- **Latent Variables:** Latent variables are unobserved variables in statistical models that can only be inferred indirectly through their effects on observable variables. They cannot be directly measured but can be detected by their impact on the observable variables.
  - Suppose we want to study why some students perform better in examinations than others.
    - Observable variables: Examination marks, problem-solving scores, and memory test scores.
    - Latent variable: Intelligence, which cannot be measured directly.
    - Inference: If a student consistently performs well across different cognitive tests, we may infer that the student has a higher level of intelligence.

- **Likelihood:** It is the probability of observing the given data given the parameters of the model. In the EM algorithm, the goal is to find the parameters that maximize the likelihood
- **Maximum Likelihood Estimation (MLE)**: MLE is a method to estimate the parameters of a statistical model by finding the parameter values that maximize the likelihood function, which measures how well the model explains the observed data.
- **Convergence:** Convergence refers to the condition when the EM algorithm has reached a stable solution. It is typically determined by checking if the change in the log-likelihood or the parameter estimates falls below a predefined threshold.

Working

- The EM algorithm works by repeatedly performing two steps:
  - E-step (Expectation): Estimate the hidden information using the current parameter values.
  - M-step (Maximization): Update the model parameters to maximize the expected likelihood of the observed data.
- These steps are repeated until the parameter estimates converge.
- The essence of Expectation-Maximization algorithm is to use the available observed data of the dataset to estimate the missing data and then using that data to update the values of the parameters.



Consider a dataset $X = \{x_1, x_2, \ldots, x_N\}$, with hidden variables Z and model parameters θ.

The likelihood of the observed data is

$$L(\theta) = p(X \mid \theta)$$

Where, ɵ represents the set of all model parameters that we need to estimate.
For example, if a Gaussian Mixture Model is considered,

$$\theta = \{\mu_k, \sigma_k^2, \pi_k\}_{k=1}^{K}$$

where:

- μ_k = mean of cluster k.
- σ_k^2 = variance of cluster k.
- π_k = mixture weight (proportion) of cluster k.

When the hidden variables are unknown, the likelihood is obtained by summing over their possible values:

$$p(X \mid \theta) = \sum_{Z} p(X, Z \mid \theta)$$

The objective is to find parameter values that maximize this likelihood.

E-step — Expectation

Using the current parameters $\theta^{(t)}$, calculate the expected complete-data log-likelihood:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Z|X,\theta^{(t)}}[\log p(X, Z \mid \theta)]$$

In simple terms, estimate the hidden information based on the current model.

M-step — Maximization

Update the parameters by maximizing the expected log-likelihood:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)})$$

The new parameters are then used in the next E-step.

- Example: Grouping student marks into two clusters

Suppose the marks of four students are:

X={10, 28,12, 30}

We suspect that the students belong to two groups:

- Cluster 1: students from batch A.
- Cluster 2: students from batch B.

However, the cluster membership of each student is unknown.

For simplicity, assume two Gaussian distributions with equal variance and equal mixture weights, and initially choose the means:

$$\mu_1 = 10, \qquad \mu_2 = 20$$

Step 1: E-step

Estimate the probability that each student belongs to each cluster.

For a Gaussian mixture model, the responsibility of cluster k for data point x_i is

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}$$

where:

- r_{ik} = probability that point x_i belongs to cluster k.
- π_k = mixture weight of cluster k.
- $\mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$ = Gaussian probability density.

For example, if the current model gives a student with 10 marks a responsibility of 0.99 for Cluster 1, the student is very likely to belong to Cluster 1.

The algorithm computes these probabilities for all students and both clusters.

Step 2: M-step

Use the responsibilities to update the cluster means.

$$\mu_k^{\text{new}} = \frac{\sum_{i=1}^{N} r_{ik} x_i}{\sum_{i=1}^{N} r_{ik}}$$

This is a **weighted mean**: data points with higher membership probabilities contribute more strongly to the new mean.

The mixture weights are updated as

$$\pi_k^{\text{new}} = \frac{1}{N} \sum_{i=1}^{N} r_{ik}$$

The variances can also be updated:

$$(\sigma_k^2)^{\text{new}} = \frac{\sum_{i=1}^{N} r_{ik} (x_i - \mu_k^{\text{new}})^2}{\sum_{i=1}^{N} r_{ik}}$$

Step 3: Repeat

The updated means, weights and variances are used in the next E-step.

After several iterations, the means will generally move toward the two groups of marks, approximately around 11 and 29 in this example, provided the model is initialized and fitted appropriately.

The final clustering is based on the estimated membership probabilities.

Usage of EM algorithm –

- It can be used to fill the missing data in a sample.
- It can be used as the basis of unsupervised learning of clusters.
- It can be used for the purpose of estimating the parameters of Hidden Markov Model (HMM).
- It can be used for discovering the values of latent variables.

- **Advantages of EM algorithm –**
  - It is always guaranteed that likelihood will increase with each iteration.
  - The E-step and M-step are often pretty easy for many problems in terms of implementation.
  - Solutions to the M-steps often exist in the closed form (completed in a finite number of steps).
- **Disadvantages of EM algorithm –**
  - It has slow convergence.
  - It makes convergence to the local optima only.

Example

Let C1 and C2 be two coins.

Θ_1 be probability of getting head with C1

Θ_2 be probability of getting head with C2

Find value of Θ_1 and Θ_2 by tossing C1 and C2 for multiple times..

[The EM Algorithm Clearly Explained (Expectation-Maximization Algorithm)]

Example

Toss for 5 times Choosing any of the coin randomly.

Each selected coin has to toss for 10 times.

B	H	T	T	T	H	H	T	H	T	H
A	H	H	H	H	T	H	H	H	H	H
A	H	T	H	H	H	H	H	T	H	H
B	H	T	H	T	T	T	H	H	T	T
A	T	H	H	H	T	H	H	H	T	H

$$\Theta_1 = \frac{\text{no of heads with C1}}{\text{Total no of flips using C1}}$$

$$\Theta_2 = \frac{\text{no of heads with C2}}{\text{Total no of flips using C2}}$$

Example

If we know the coin labels the probability will be as follows:

Coin A	Coin B
	5 H, 5 T
9 H, 1 T	
8 H, 2 T	
	4 H, 6 T
7 H, 3 T	

$$\Theta_1 = 24/(24+6) = 0.8$$

$$\Theta_2 = 9/(9+11) = 0.45$$

[EM Algorithm with Example]

Example

If we don't know the identity of coin labels then we will assume or estimate the probabilities.

$\Theta_1 = 0.6$

$\Theta_2 = 0.5$

We have to use binomial distribution to find likelihood.

$L(C)=\Theta^k (1-\Theta)^{n-k}$

likelihood function for a sequence of Bernoulli trials (like coin tosses).

- $\theta \rightarrow$ Probability of success (e.g., getting heads)
- $k \rightarrow$ Number of successes observed
- $n \rightarrow$ Total number of trials
- $n - k \rightarrow$ Number of failures

Why Binomial distribution is used here?
2 outcomes; H and T only

Bring in **Bayes' Theorem**

$$P(\theta \mid D) = \frac{P(D|\theta)\, P(\theta)}{P(D)}$$

- D = our data (k successes in n trials)
- P(D|θ) = likelihood (given formula)
- P(θ) = prior belief about θ
- P(θ|D) = posterior (updated belief)

$$P(\theta \mid D) \propto \theta^k (1-\theta)^{n-k} \cdot P(\theta)$$

Likelihood × Prior → Posterior

here, n=no. of flips taken by the coin and k= the number of heads

[Numerical example: The EM Algorithm Clearly Explained (Expectation-Maximization Algorithm)]

Example

$$P(A \mid k) = \frac{L(A) \cdot \pi}{L(A) \cdot \pi + L(B) \cdot (1 - \pi)}$$

$$L(C) = \Theta^{k} (1 - \Theta)^{n-k}$$

$$\frac{L(A)}{L(A) + L(B)}$$

Likelihood For first coin Flips

$L(A) = 0.6^{5} (1 - 0.6)^{10-5} = 0.0007963$

$L(B) = 0.5^{5} (1 - 0.5)^{10-5} = 0.0009766$

$P(A) = L(A)/L(A)+L(B) = 0.0007963/(0.0007963+0.0009766) = 0.45$

$P(B) = L(B)/L(A)+L(B) = 0.0009766/(0.0007963+0.0009766) = 0.55$

Example

In similar fashion find probability of all coins with all flips. It will be as follows:

L(H): Likely no of heads L(T): Likely no of tails

											Iteration 1->:		Coin A		Coin B	
											P(A)	P(B)	L(H)	L(T)	L(H)	L(T)
B	H	T	T	T	H	H	T	H	T	H	0.45	0.55	2.2	2.2	2.8	2.8
A	H	H	H	H	T	H	H	H	H	H	0.80	0.20	7.2	0.8	1.8	0.2
A	H	T	H	H	H	H	H	T	H	H	0.73	0.27	5.9	1.5	2.1	0.5
B	H	T	H	T	T	T	H	H	T	T	0.35	0.65	1.4	2.1	2.6	3.9
A	T	H	H	H	T	H	H	H	T	H	0.65	0.35	4.5	1.9	2.5	1.1

RECORDED WITH SCREENCAST-O-MATIC

Example

For Coin A:

$\sum L(H) = 21.3$

$\sum L(T) = 8.6$

$\Theta_1 = 21.3/(21.3+8.6)$

$= 0.71$

For Coin B:

$\sum L(H) = 11.7$

$\sum L(T) = 8.4$

$\Theta_2 = 11.7/(11.7+8.4)$

$= 0.58$

These values of Θ_1 and Θ_2 will be sent to next iteration.

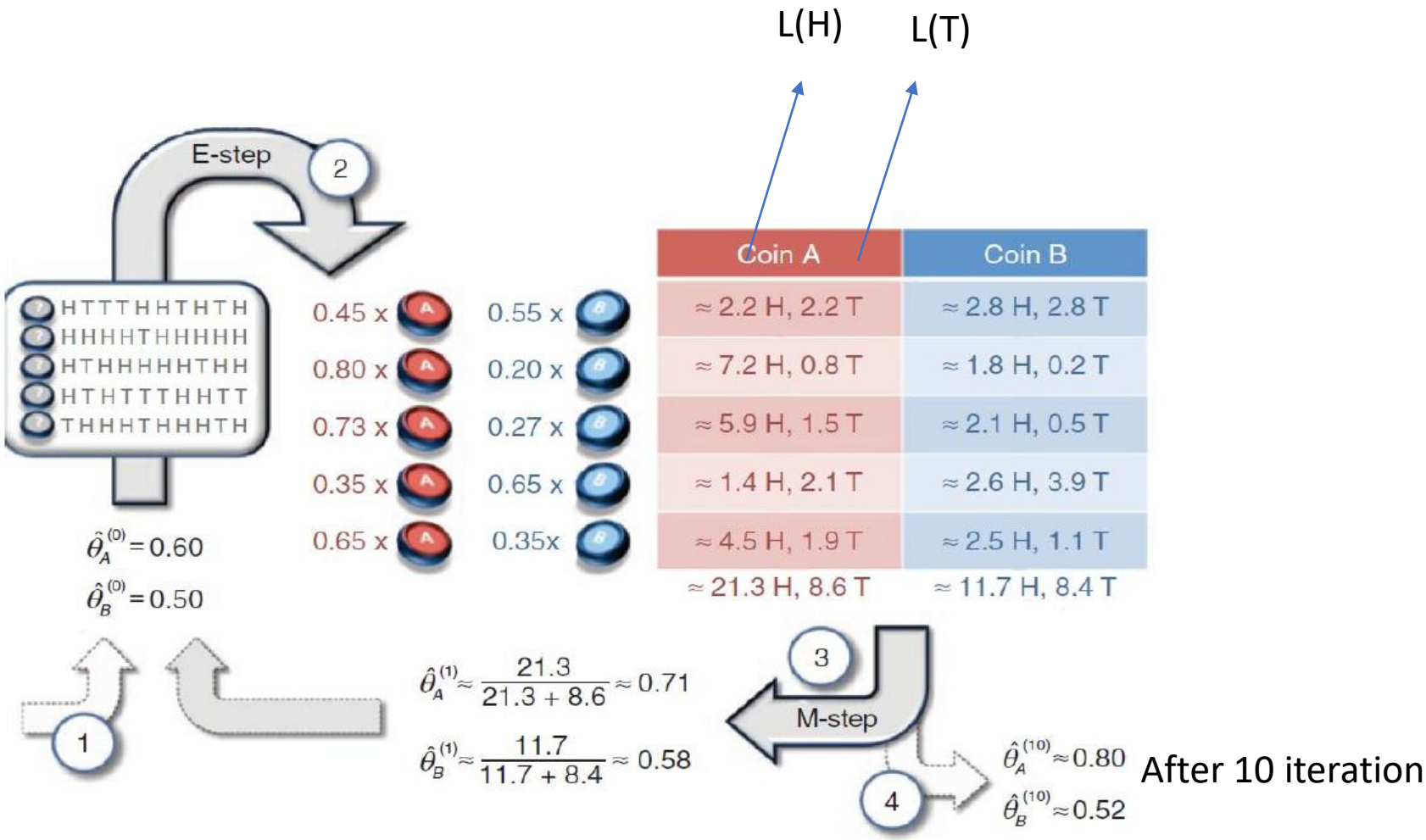
After 10 iteration:

$\Theta = 0.80$ and $\Theta = 0.52$

The process will be continued until you get stable value of Θ_1 and Θ_2.

RECORDED WITH SCREENCAST MATIC

- Summary



After 10 iteration

Example2: Soft Clustering with EM

Estimation: Estimate the expectation from machine on some data to cluster

Maximization: Whatever is estimated should be maximized to find best result.

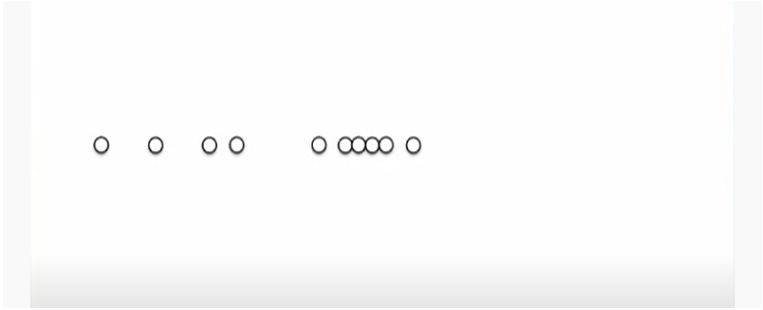
It starts with random data and repeats two steps till best result.

E Step: Cluster based on current data

M Step: Generate best theory to get best clusters.

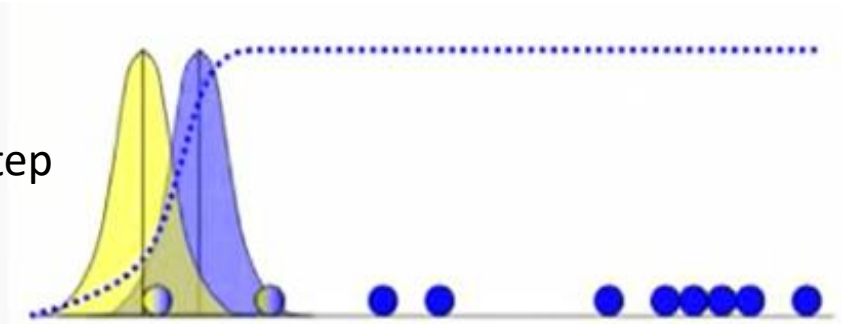
EM algorithm: how it works
Expectation Maximization: how it works

Example 2: Probability of data points in cluster A or B



Data Points

E-Step



$$P(x_i \mid b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{(x_i - \mu_b)^2}{2\sigma_b^2}\right)$$

$$b_i = P(b \mid x_i) = \frac{P(x_i \mid b)P(b)}{P(x_i \mid b)P(b) + P(x_i \mid a)P(a)}$$

$$a_i = P(a \mid x_i) = 1 - b_i$$

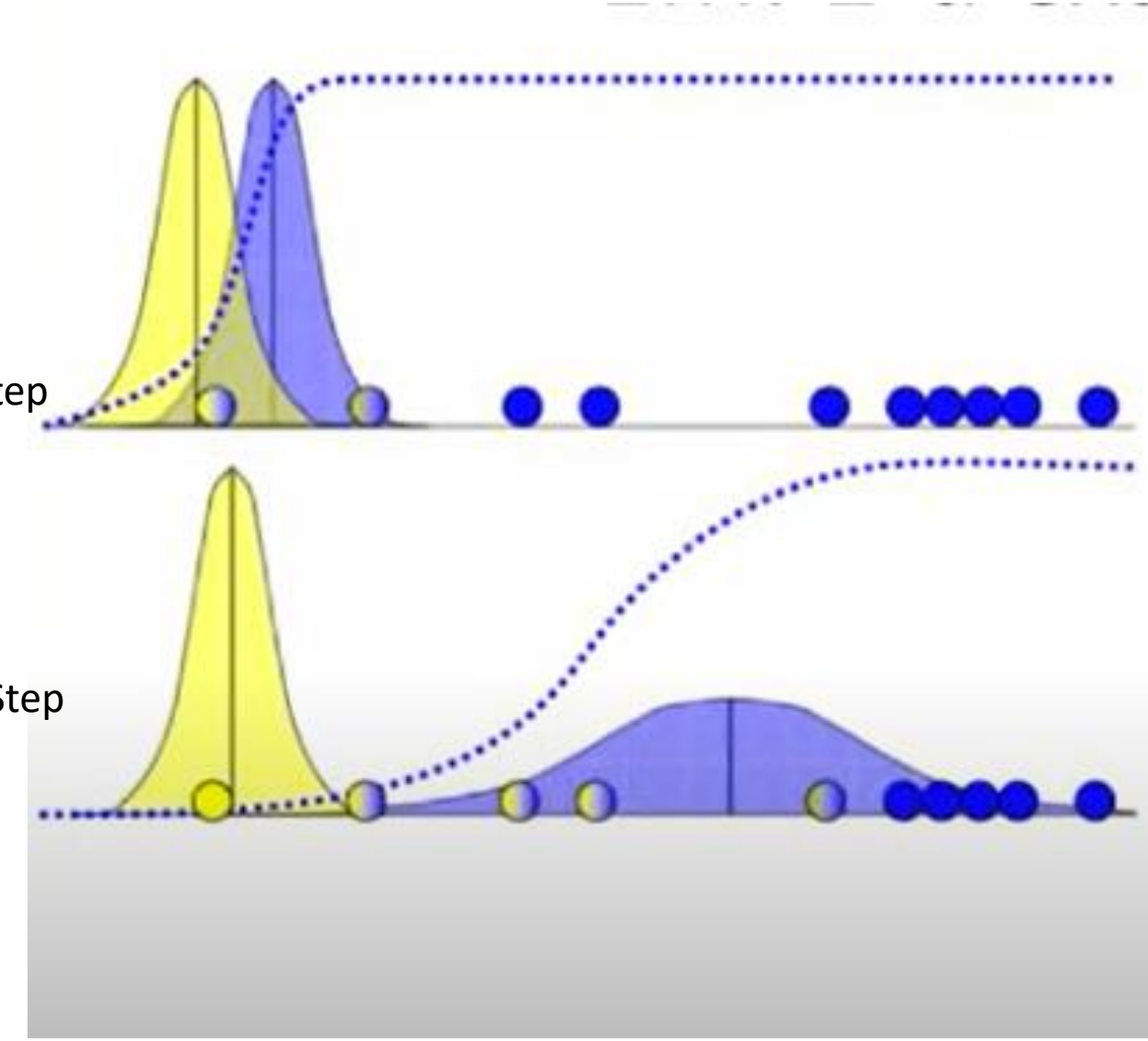
$$P(b) = (b_1 + b_2 + \ldots b_n) / n$$
$$P(a) = 1 - P(b)$$

Expectation Step

Maximization Step

E-Step

M-Step



$$P(x_i \mid b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{(x_i - \mu_b)^2}{2\sigma_b^2}\right)$$

$$b_i = P(b \mid x_i) = \frac{P(x_i \mid b)P(b)}{P(x_i \mid b)P(b) + P(x_i \mid a)P(a)}$$

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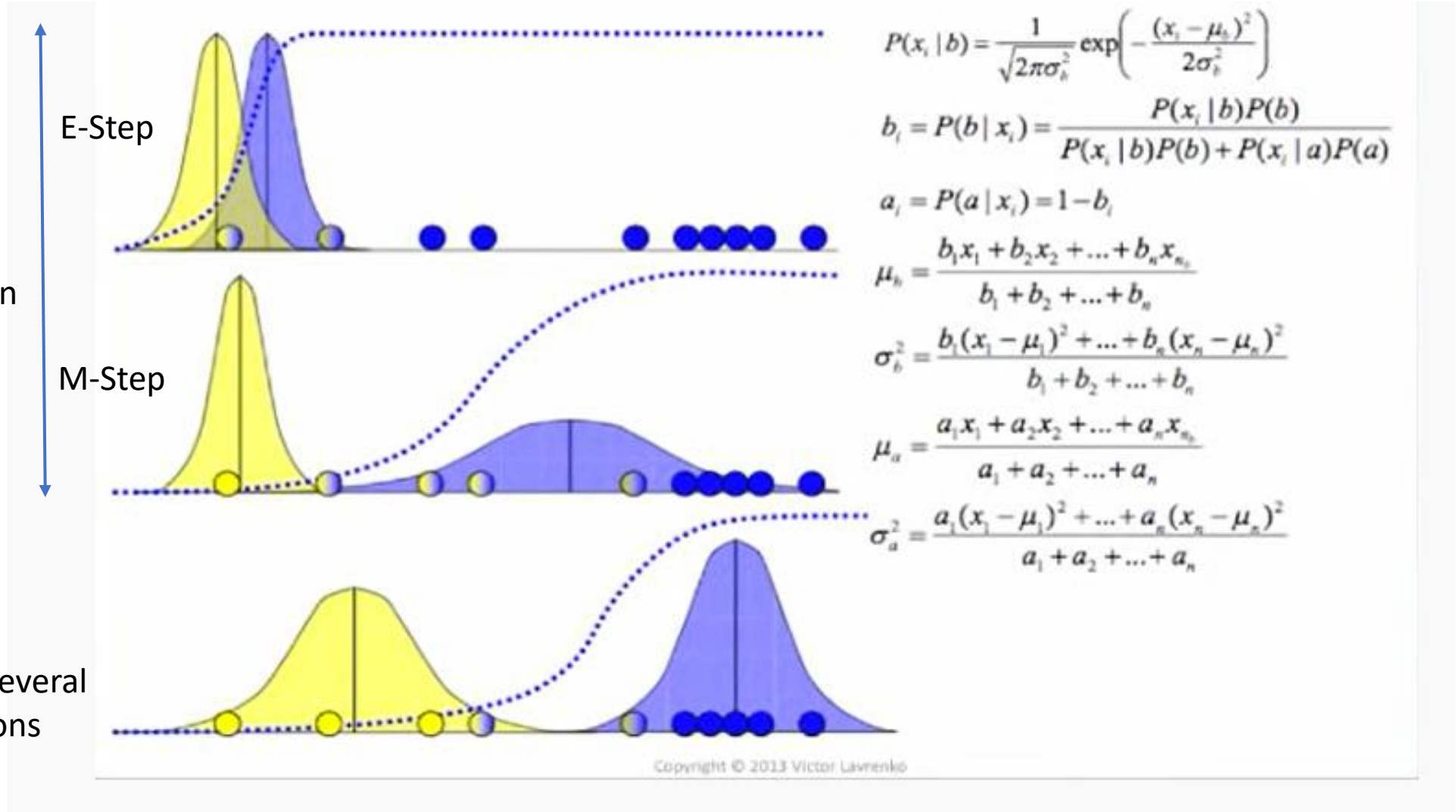
$$\mu_b = \frac{b_1 x_1 + b_2 x_2 + \ldots + b_n x_{n_b}}{b_1 + b_2 + \ldots + b_n}$$

$$\sigma_b^2 = \frac{b_1 (x_1 - \mu_1)^2 + \ldots + b_n (x_n - \mu_n)^2}{b_1 + b_2 + \ldots + b_n}$$

$$\mu_a = \frac{a_1 x_1 + a_2 x_2 + \ldots + a_n x_{n_a}}{a_1 + a_2 + \ldots + a_n}$$

$$\sigma_a^2 = \frac{a_1 (x_1 - \mu_1)^2 + \ldots + a_n (x_n - \mu_n)^2}{a_1 + a_2 + \ldots + a_n}$$

- Summary



$$P(x_i \mid b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{(x_i - \mu_b)^2}{2\sigma_b^2}\right)$$

$$b_i = P(b \mid x_i) = \frac{P(x_i \mid b)P(b)}{P(x_i \mid b)P(b) + P(x_i \mid a)P(a)}$$

$$a_i = P(a \mid x_i) = 1 - b_i$$

$$\mu_b = \frac{b_1 x_1 + b_2 x_2 + \ldots + b_n x_{n_b}}{b_1 + b_2 + \ldots + b_n}$$

$$\sigma_b^2 = \frac{b_1 (x_1 - \mu_1)^2 + \ldots + b_n (x_n - \mu_n)^2}{b_1 + b_2 + \ldots + b_n}$$

$$\mu_a = \frac{a_1 x_1 + a_2 x_2 + \ldots + a_n x_{n_a}}{a_1 + a_2 + \ldots + a_n}$$

$$\sigma_a^2 = \frac{a_1 (x_1 - \mu_1)^2 + \ldots + a_n (x_n - \mu_n)^2}{a_1 + a_2 + \ldots + a_n}$$

I[st] Iteration

After several Iterations

Appendix

Basics of Gaussian Distribution

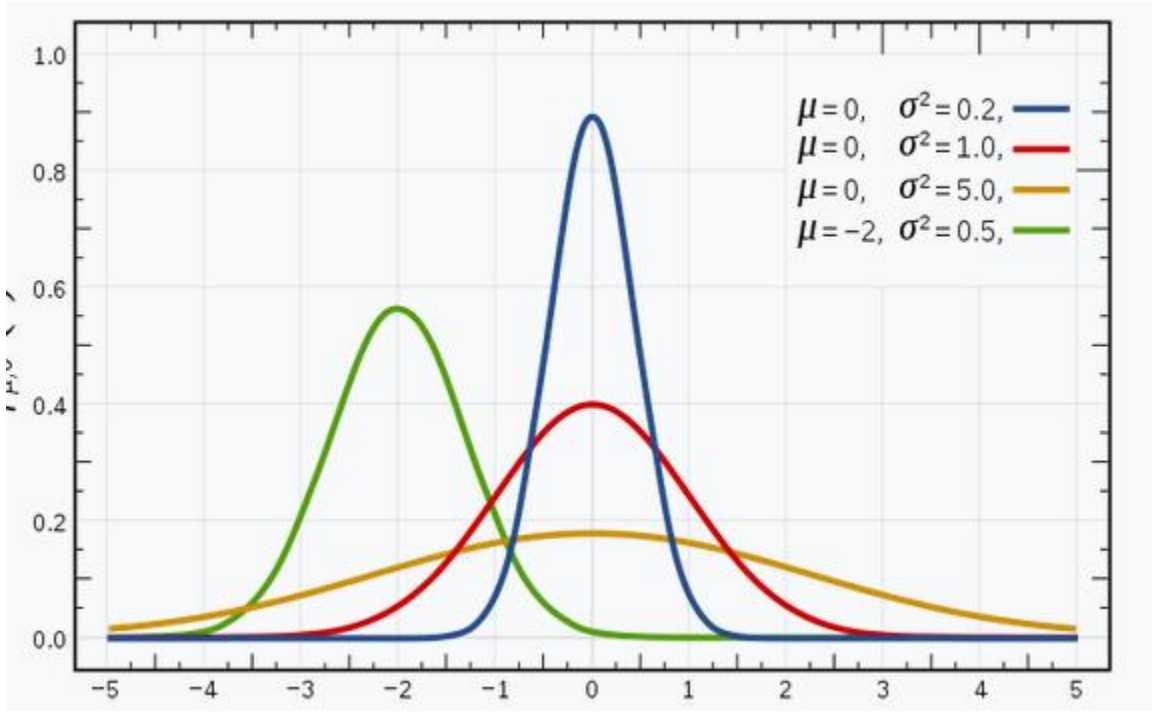
- Gaussian Distributions (or the Normal Distribution) - It has a bell-shaped curve, with the observations symmetrically distributed around the mean (average) value.
- The given image shown has a few Gaussian distributions with different values of the mean (μ) and variance (σ^2).
- In 1-D space, the probability density function of a Gaussian distribution is given by:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ represents the mean

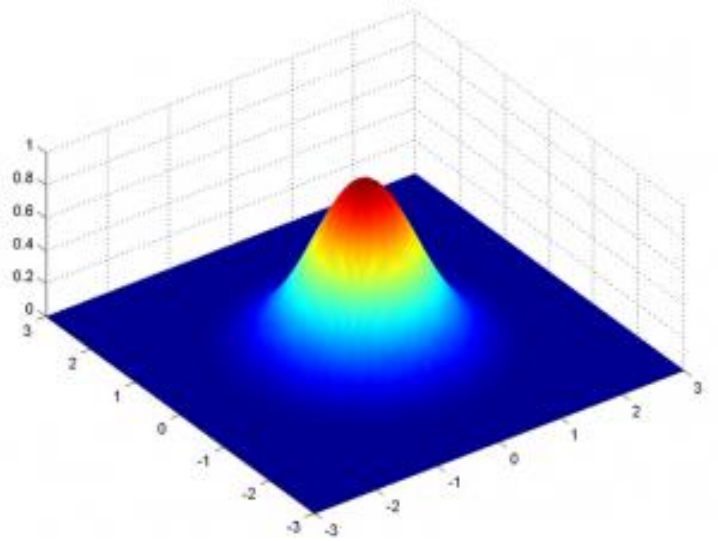
  and σ^2 represents the variance.

[The Normal Distribution, Clearly Explained!!! - YouTube]



- In the case of two variables, we will have a 3D bell curve instead of a 2D bell-shaped curve as shown below:
- The probability density function would be given by:

$$f(x \mid \mu, \Sigma) = \frac{1}{\sqrt{2\pi|\Sigma|}} \exp\left[-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)\right]$$



- μ is the 2-D mean vector,

  and Σ is the 2×2 covariance matrix.

We can generalize the same for the d-dimension.

- Thus, for the Multivariate Gaussian model, we have x and μ as vectors of length d, and Σ would be a *d x d* covariance matrix.
- Hence, for a dataset having *d* features, we would have a mixture of *k* Gaussian distributions (where *k* represents the number of clusters), each having a certain mean vector and variance matrix.
- The question is: **" How we can find out the mean and variance for each Gaussian?"**
- For finding these values, we using a technique called Expectation-Maximization (EM).

Gaussian Mixture Models

- The main assumption of these mixture models is that there are a certain number of Gaussian distributions, and each of these distributions represents a cluster.
- Hence, a Gaussian Mixture model tries to group the observations belonging to a single distribution together.
- **For Example,** the Gaussian Mixture Model of 2 Gaussian distributions
- We have two Gaussian distributions- **$N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$**
- Here, we have to estimate a total of 5 parameters:
- **θ = (p, μ_1, σ_1^2,μ_2, σ_2^2)**
- where p is the probability that the data comes from the first Gaussian distribution and 1-p that it comes from the second Gaussian distribution.

- **Then, the probability density function (PDF) of the mixture model is given by:**
- **$g(x|\theta) = p\, g_1(x|\, \mu_1, \sigma_1^2) + (1\text{-}p) g_2(x|\, \mu_2, \sigma_2^2)$**
- **Objective:** To best fit a given probability density by finding

  $\theta = (\, p, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2)$ through EM iterations.

➢What is the probability of observing a datapoint with value x?

$$p(x;\theta) = \pi\phi(x;\mu_1,\sigma_1) + (1-\pi)\phi(x;\mu_2,\sigma_2)$$

- The probability (likelihood) of observing entire dataset of *n* points is:

$$L(\theta;\text{data}) = \prod_{i=1}^{n} p(x_i;\theta)$$

- And we typically choose to take the logarithm of this to turn our product into a more manageable sum, the log-likelihood

$$\ell(\theta; \text{data}) = \sum_{i=1}^{n} \log p(x_i; \theta)$$

$$= \sum_{i=1}^{n} \log \left[\pi \phi(x_i; \mu_1, \sigma_1) + (1 - \pi)\phi(x_i; \mu_2, \sigma_2)\right]$$

- Our goal is to maximize this: we want our parameters to be the ones where it is most likely that we observe the data we observed (a Maximum Likelihood Estimator).

Some basics

- Probability distributions and EM
  - The Normal Distribution, Clearly Explained!!! – YouTube
  - https://www.youtube.com/watch?v=3Ij1lIsC1Xg
  - EM.1: Introduction to mixture models – YouTube
  - EM.2: Expectation-maximization algorithm – YouTube
  - EM.3: Visualizing the EM algorithm – YouTube
  - EM.4: Gaussian mixture model (GMM) – YouTube
  - EM.5: How many Gaussians in the GMM? – YouTube
  - EM.6: Summary - YouTube